Project Task 5

Programme: Bachelor of Computer Science (Data Engineering)

Subject: SECR1213 Network Communications

Session-Sem: 2024/2025-1

Section: 02

Group name: Router ranger

Group Member	Matric Number
Choh Jing Yi	A23CS0296
Neo Li Xin	A23CS0253
Pravinraj a/l Sivabathi	A23CS0171
Ahmad Ziyaad bin Mohd Abbas	A23CS0206

Index

5.1 Get the Network Address from your lecturer

Group/Section	Network Address
1	10.16.0.0/12
2	10.32.0.0/12
3	10.48.0.0/12
4	10.64.0.0/12
5	10.80.0.0/12
6	10.96.0.0/12
7	10.112.0.0/12
8	10.128.0.0/12
9	10.144.0.0/12
10	10.176.0.0/12

Table 5.1.1

Table 5.1.1 shows the list of network addresses of each group. The network address of our group is 10.128.0.0/12.

Network address details:

IP address (decimal)	10.128.0.0
IP address (binary)	00001010 10000000 00000000 00000000
Subnet Mask(decimal)	255.240.0.0
Subnet Mask(binary)	11111111 11110000 00000000 00000000

We partitioned the subnet mask into a network portion (12 bits) and a host portion (20 bits). By performing the AND operation of the IP address with the subnet mask, we identified the network address as 10.128.0.0 and the broadcast address as 10.143.255.255. The valid address range within our building spans from 10.128.0.1 to 10.143.255.254.

Network Address	10.128.0.0
Broadcast Address	10.143.255.255
Range of usable address	10.128.0.1 - 10.143.255.254

5.2 Divide it in the best possible way for your network.

5.2.1 Network distribution for different work area

In order to improve internet speed and enhance security of the faculty work areas, subnetting is done to divide the IP address space into fixed-size classes. Based on the network design requirements, we identified that we need 8 subnets to accommodate 10 different work areas, where some subnets are used to manage several work areas. To achieve these, 3 extra bits (2^3=8) is borrowed from host portion to construct subnet IP address. The table below listed each subnet network portion and host portion and the distribution of subnet for each area.

Subnet	**Network portion**	**Subnet bits**	**Host portion**		**IP address**
0	00001010 1000	000	0 0000 0000 0000 0000	Network Address	10.128.0.0
			1 1111 1111 1111 1111	Broadcast Address	10.129.255.255
1	00001010 1000	001	0 0000 0000 0000 0000	Network Address	10.130.0.0
			1 1111 1111 1111 1111	Broadcast Address	10.131.255.255
2	00001010 1000	010	0 0000 0000 0000 0000	Network Address	10.132.0.0
			1 1111 1111 1111 1111	Broadcast Address	10.133.255.255
3	00001010 1000	011	0 0000 0000 0000 0000	Network Address	10.134.0.0
			1 1111 1111 1111 1111	Broadcast Address	10.135.255.255
4	00001010 1000	100	0 0000 0000 0000 0000	Network Address	10.136.0.0

			1 1111 1111 1111 1111	Broadcast Address	10.137.255.255
5	00001010 1000	101	0 0000 0000 0000 0000	Network Address	10.138.0.0
			1 1111 1111 1111 1111	Broadcast Address	10.139.255.255
6	00001010 1000	110	0 0000 0000 0000 0000	Network Address	10.140.0.0
			1 1111 1111 1111 1111	Broadcast Address	10.141.255.255
7	00001010 1000	111	0 0000 0000 0000 0000	Network Address	10.142.0.0
			1 1111 1111 1111 1111	Broadcast Address	10.143.255.255

Distribution of subnet for different work area:

Subnet	**Work Area**	**Network Address**	**Broadcast Address**	**Range of usable address**	**Subnet Mask Address**
0	Server Room	10.128.0.0	10.129.255.255	10.128.0.1 - 10.129.255.254	/15 255.254.0.0
First Floor					
1	General Lab 1	10.130.0.0	10.131.255.255	10.130.0.1 - 10.131.255.254	/15 255.254.0.0

2	General Lab 2	10.132.0.0	10.133.255.255	10.132.0.1 - 10.133.255.254	/15 255.254.0.0
3	Student Lounge and Student Meeting Room	10.134.0.0	10.135.255.255	10.134.0.1 - 10.135.255.254	/15 255.254.0.0
Second Floor					
4	Office & Video Conference Room	10.136.0.0	10.137.255.255	10.136.0.1 - 10.137.255.254	/15 255.254.0.0
5	Cisco Network Lab	10.138.0.0	10.139.255.255	10.138.0.1 - 10.139.255.254	/15 255.254.0.0
6	Embedded Room	10.140.0.0	10.141.255.255	10.140.0.1 - 10.141.255.254	/15 255.254.0.0
7	Hybrid Classroom	10.142.0.0	10.143.255.255	10.142.0.1 - 10.143.255.254	/15 255.254.0.0

5.2.2 Network distribution for workstation in different work areas.

First Floor:

1. General Lab 1

Devices	IP Address
PC1	10.130.0.1
PC2	10.130.0.2
PC3	10.130.0.3
PC4	10.130.0.4
PC5	10.130.0.5
PC6	10.130.0.6
PC7	10.130.0.7
PC8	10.130.0.8
PC9	10.130.0.9
PC10	10.130.0.10
PC11	10.130.0.11
PC12	10.130.0.12
PC13	10.130.0.13
PC14	10.130.0.14
PC15	10.130.0.15
PC16	10.130.0.16
PC17	10.130.0.17
PC18	10.130.0.18
PC19	10.130.0.19
PC20	10.130.0.20
PC21	10.130.0.21
PC22	10.130.0.22
PC23	10.130.0.23
PC24	10.130.0.24
PC25	10.130.0.25
PC26	10.130.0.26
PC27	10.130.0.27
PC28	10.130.0.28
PC29	10.130.0.29
PC30	10.130.0.30
PC31	10.130.0.31
PC32	10.130.0.32
PJ1	10.130.0.33
WAP1	10.130.0.34

2. General Lab 2

Devices	IP Address
PC33	10.132.0.1

PC34	10.132.0.2
PC35	10.132.0.3
PC36	10.132.0.4
PC37	10.132.0.5
PC38	10.132.0.6
PC39	10.132.0.7
PC40	10.132.0.8
PC41	10.132.0.9
PC42	10.132.0.10
PC43	10.132.0.11
PC44	10.132.0.12
PC45	10.132.0.13
PC46	10.132.0.14
PC47	10.132.0.15
PC48	10.132.0.16
PC49	10.132.0.17
PC50	10.132.0.18
PC51	10.132.0.19
PC52	10.132.0.20
PC53	10.132.0.21
PC54	10.132.0.22
PC55	10.132.0.23
PC56	10.132.0.24
PC57	10.132.0.25
PC58	10.132.0.26
PC59	10.132.0.27
PC60	10.132.0.28
PC61	10.132.0.29
PC62	10.132.0.30
PC63	10.132.0.31
PC64	10.132.0.32
PJ2	10.132.0.33
WAP2	10.132.0.34

3. Student Lounge

Devices	IP Address
PC65	10.134.0.1
PC66	10.134.0.2
PC67	10.134.0.3
PC68	10.134.0.4
PC69	10.134.0.5
PC70	10.134.0.6
PC71	10.134.0.7

PC72	10.134.0.8
PRINTER1	10.134.0.9
WAP3	10.134.0.10

4. Student Meeting Room

Devices	IP Address
PJ4	10.135.0.1
MONITOR2	10.135.0.2
WAP4	10.135.0.3

Second floor:

5. Office

Devices	IP Address
PC73	10.136.0.1
PC74	10.136.0.2
PC75	10.136.0.3
PC76	10.136.0.4
PC77	10.136.0.5
PC78	10.136.0.6
PC79	10.136.0.7
PC80	10.136.0.8
WAP5	10.136.0.9

6. Video Conference Room

Devices	IP Address
PJ5	10.137.0.1
MONITOR	10.137.0.2
WAP6	10.137.0.3

7. Cisco Network Lab

Devices	IP Address
PC81	10.138.0.1
PC82	10.138.0.2
PC83	10.138.0.3
PC84	10.138.0.4
PC85	10.138.0.5
PC86	10.138.0.6
PC87	10.138.0.7
PC88	10.138.0.8
PC89	10.138.0.9
PC90	10.138.0.10
PC91	10.138.0.11
PC92	10.138.0.12
PC93	10.138.0.13
PC94	10.138.0.14
PC95	10.138.0.15
PC96	10.138.0.16
PC97	10.138.0.17
PC98	10.138.0.18
PC99	10.138.0.19
PC100	10.138.0.20
PC101	10.138.0.21
PC102	10.138.0.22
PC103	10.138.0.23
PC104	10.138.0.24
PC105	10.138.0.25
PC106	10.138.0.26
PC107	10.138.0.27
PC108	10.138.0.28
PC109	10.138.0.29
PC110	10.138.0.30
PC111	10.138.0.31
PC112	10.138.0.32
SWITCH1	10.138.0.33
SWITCH2	10.138.0.34
SWITCH3	10.138.0.35
SWITCH4	10.138.0.36
SWITCH5	10.138.0.37
SWITCH6	10.138.0.38
SWITCH7	10.138.0.39
SWITCH8	10.138.0.40
SWITCH9	10.138.0.41
SWITCH10	10.138.0.42

SWITCH11	10.138.0.43
SWITCH12	10.138.0.44
SWITCH13	10.138.0.45
SWITCH14	10.138.0.46
SWITCH15	10.138.0.47
SWITCH16	10.138.0.48
ROUTER1	10.138.0.49
ROUTER2	10.138.0.50
ROUTER3	10.138.0.51
ROUTER4	10.138.0.52
ROUTER5	10.138.0.53
ROUTER6	10.138.0.54
ROUTER7	10.138.0.55
ROUTER8	10.138.0.56
ROUTER9	10.138.0.57
ROUTER10	10.138.0.58
ROUTER11	10.138.0.59
ROUTER12	10.138.0.60
ROUTER13	10.138.0.61
ROUTER14	10.138.0.62
ROUTER15	10.138.0.63
ROUTER16	10.138.0.64
ROUTER17	10.138.0.65
ROUTER18	10.138.0.66
ROUTER19	10.138.0.67
ROUTER20	10.138.0.68
ROUTER21	10.138.0.69
ROUTER22	10.138.0.70
ROUTER23	10.138.0.71
ROUTER24	10.138.0.72
PJ6	10.138.0.73
WAP7	10.137.0.74

8. Embedded Room

Devices	IP Address
PC113	10.140.0.1
PC114	10.140.0.2
PC115	10.140.0.3
PC116	10.140.0.4
PC117	10.140.0.5
PC118	10.140.0.6
PC119	10.140.0.7
PC120	10.140.0.8

PC121	10.140.0.9
PC122	10.140.0.10
PC123	10.140.0.11
PC124	10.140.0.12
PC125	10.140.0.13
PC126	10.140.0.14
PC127	10.140.0.15
PC128	10.140.0.16
PC129	10.140.0.17
PC130	10.140.0.18
PC131	10.140.0.19
PC132	10.140.0.20
PC133	10.140.0.21
PC134	10.140.0.22
PC135	10.140.0.23
PC136	10.140.0.24
PC137	10.140.0.25
PC138	10.140.0.26
PC139	10.140.0.27
PC140	10.140.0.28
PC141	10.140.0.29
PC142	10.140.0.30
PJ7	10.140.0.31
WEBCAM	10.140.0.32
SCANNER	10.140.0.33
SMARTLOCK	10.140.0.34
MONITOR2	10.140.0.35
PRINTER2	10.140.0.36
WAP8	10.140.0.37

9. Hybrid Classroom

Devices	IP Address
PJ7	10.142.0.1
MONITOR3	10.142.0.2
MONITOR4	10.142.0.3
PROJECTOR	10.142.0.4
WAP9	10.142.0.5

10. Server Room

Devices	IP Address
SV1	10.128.0.1
SV2	10.128.0.2

SV3	10.128.0.3
SV4	10.128.0.4
SV5	10.128.0.5
SV6	10.128.0.6
SV7	10.128.0.7
SV8	10.128.0.8
SV9	10.128.0.9
SV10	10.128.0.10

5.3 Meeting Minutes

Meeting Minutes #1

Date/Time	24 December 2024 10:00pm
Location Online	Google Meet
Agenda	1. Overviewing task 2. Task distribution 3. Completing Task 5
Meeting MC	Choh Jing Yi

Attendance

Name	Time	Reason For Absence
Choh Jing Yi	22 00	-
Neo Li Xin	22 00	-
Pravinraj a/l Sivabathi	22 00	-
Ahmad Ziyaad bin Mohd Abbas	22 00	-

Meeting details

NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discussion for Task 5	- Choh Jing Yi shared the information about task 5 and its rubric. -All members read and investigate the task carefully. -Solved the problem raised by group members together through discussion.	All members
2	Suggestion for subnetwork division	- All members suggest their prefer subnetwork division - All members discuss and decide the one that best suits our design.	All members
3	IP Addressing Scheme overview	- All members examine how IP addresses are assigned to each work area.	All members
4	Proceed to the parts of Task 5	-All members work on Microsoft Word together to complete task 5.	All members

		-At the same time, if any of the group members question and identify any issues or problems then we find ways to encounter problems together.	
5	Meeting Ended	-At 12.00pm, the meeting ended after all the discussion and work is done.	All members